📘 SQL CLASS NOTE – MODULE 10: `LIKE`

1. Introduction

The `LIKE` **operator** is used in SQL to **search for a pattern in a column's value**.
It is mostly used with **text columns** (e.g., names, grades, courses).

👉 `LIKE` works with **wildcards**:

- `%` → Matches **zero, one, or many characters**
- `_` → Matches **exactly one character**

2. Syntax

```sql
SELECT column1, column2, ...
FROM table_name
WHERE column_name LIKE 'pattern';
```

3. Wildcard Patterns

Pattern	Meaning	Example
`'A%'`	Starts with A	"Alice", "Andrew"
`'%son'`	Ends with "son"	"Jackson", "Emerson"
`'%an%'`	Contains "an"	"Hannah", "Jonathan"
`'_a%'`	Second letter is "a"	"Mary", "David"
`'____'`	Exactly 4 letters	"John", "Mark"

4. Practical Examples with `SchoolDB`

Example 1: Find students whose first name starts with "A"

```
SELECT FirstName, LastName
FROM Students
WHERE FirstName LIKE 'A%';
```

Example 2: Find students whose last name ends with "son"

```
SELECT FirstName, LastName
FROM Students
WHERE LastName LIKE '%son';
```

Example 3: Find courses that contain "Math"

```
SELECT CourseName, CourseCode
FROM Courses
WHERE CourseName LIKE '%Math%';
```

Example 4: Find students whose first name has "a" as the second letter

```
SELECT FirstName, LastName
FROM Students
WHERE FirstName LIKE '_a%';
```

Example 5: Find all students with exactly 5-letter first names

```
SELECT FirstName, LastName
FROM Students
WHERE FirstName LIKE '_____';
```

5. Why Use `LIKE` ?

- Useful for **searching text** when you don't know the exact spelling.
- Works well for **search filters** in applications.
- Helps in **data cleaning** (finding patterns, e.g., wrong emails).

6. Real-Life Use Cases

- **Schools**: Find students whose names start with "J".
- **E-commerce**: Search products containing the word "Phone".
- **Hospitals**: Find patients with diagnoses containing "Flu".
- **HR Systems**: Search employees with emails ending in "@gmail.com".

7. Classwork (Practice) ✅

👉 Try these queries:

1. Find all students whose **first names start with J**.
2. Show all students whose **last names end with "n"**.
3. Display all courses that **contain the word "Science"**.
4. Find students whose **first names contain "an"**.
5. Show all students whose **first names have exactly 4 letters**.
6. Find students whose **last name starts with "O" and ends with "a"**.
7. Find all students whose **email contains "school.com"**.

8. Challenge Question 💡

👉 Write a query to find the `FullName` , `Age` , and `GradeLevel` of students:

- Whose **first name starts with "M"**
- AND whose **last name contains "son"**

✅ That's the **expanded Module 10: LIKE**.

Perfect 👍 Let's solve the **Module 10 Classwork + Challenge** step by step with SQL queries.

✅ Solutions to Module 10: `LIKE`

1. Find all students whose **first names start with J.**

```
SELECT FirstName, LastName, GradeLevel
FROM Students
WHERE FirstName LIKE 'J%';
```

2. Show all students whose **last names end with "n".**

```
SELECT FirstName, LastName
FROM Students
WHERE LastName LIKE '%n';
```

3. Display all courses that **contain the word "Science".**

```
SELECT CourseName, CourseCode, Teacher
FROM Courses
WHERE CourseName LIKE '%Science%';
```

4. Find students whose **first names contain “an”**.

```sql
SELECT FirstName, LastName
FROM Students
WHERE FirstName LIKE '%an%';
```

5. Show all students whose **first names have exactly 4 letters.**

```sql
SELECT FirstName, LastName
FROM Students
WHERE FirstName LIKE '____';
```

6. Find students whose **last name starts with “O” and ends with “a”.**

```sql
SELECT FirstName, LastName
FROM Students
WHERE LastName LIKE 'O%a';
```

7. Find all students whose **email contains “school.com”.**

```sql
SELECT FirstName, LastName, Email
FROM Students
WHERE Email LIKE '%school.com%';
```

💡 Challenge Question

👉 Find the `FullName` , `Age` , and `GradeLevel` of students:

- Whose **first name starts with “M”**
- AND whose **last name contains “son”**

```
SELECT CONCAT(FirstName, ' ', LastName) AS FullName,
       Age,
       GradeLevel
FROM Students
WHERE FirstName LIKE 'M%'
  AND LastName LIKE '%son%';
```

📌 This query combines **two LIKE conditions** for pattern matching.

✅ That completes **Module 10: LIKE**.

👉 Should I now continue with **Module 11: ORDER BY** in the same expanded, practical style?